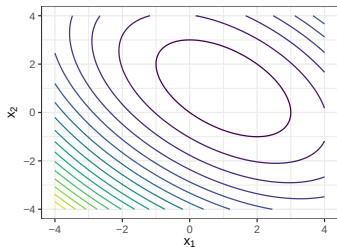
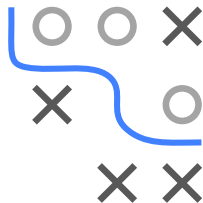


Optimization in Machine Learning

First order methods

GD on quadratic forms



Learning goals

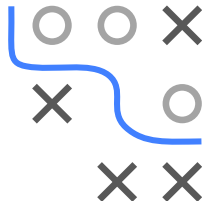
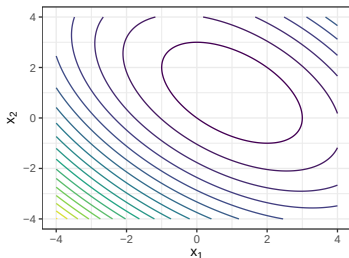
- Eigendecomposition of quadratic forms
- GD steps in eigenspace

Slides based on [► Goh 2017](#)

QUADRATIC FORMS & GD

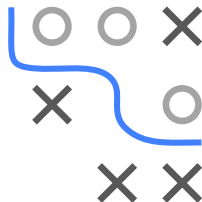
- We consider the quadratic function $q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} - \mathbf{b}^T \mathbf{x}$
- We assume that the Hessian $\mathbf{H} = 2\mathbf{A}$ has full rank
- We assume q is convex, so $\mathbf{A}$ is psd
- The optimal solution is $\mathbf{x}^* = \frac{1}{2} \mathbf{A}^{-1} \mathbf{b}$
- With $\nabla q(\mathbf{x}) = 2\mathbf{A} \mathbf{x} - \mathbf{b}$, gradient descent iterates as

$$\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} - \alpha(2\mathbf{A}\mathbf{x}^{[t]} - \mathbf{b})$$



- 
- A contour plot of a 2D Gaussian distribution in the w_1 - w_2 plane. The horizontal axis is labeled w_1 and ranges from -4 to 4. The vertical axis is labeled w_2 and ranges from -4 to 4. The plot shows several concentric elliptical contours centered at the origin (0,0). The ellipses are elongated along the w_1 axis. The contours are colored with a gradient: the innermost ellipse is dark purple, and as the ellipses move outwards, the color transitions through blue and green to yellow at the outermost edges. The background of the plot is a light gray grid.

GD STEPS IN EIGENSPACE



- With $\mathbf{w}^{[t]} = \mathbf{V}^T(\mathbf{x}^{[t]} - \mathbf{x}^*)$, a single GD step starts with

$$\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} - \alpha(2\mathbf{A}\mathbf{x}^{[t]} - \mathbf{b})$$

- This becomes

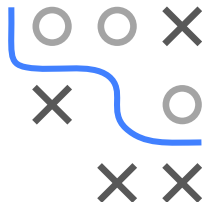
$$\mathbf{w}^{[t+1]} = \mathbf{w}^{[t]} - 2\alpha\mathbf{\Lambda}\mathbf{w}^{[t]}$$

(proof on next slide)

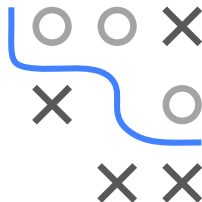
GD STEPS IN EIGENSPACE: PROOF

- A single GD step satisfies $\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} - \alpha(2\mathbf{A}\mathbf{x}^{[t]} - \mathbf{b})$
- Multiplying with $\mathbf{V}^T$ and subtracting $\mathbf{x}^*$ yields

$$\begin{aligned}\mathbf{V}^T(\mathbf{x}^{[t+1]} - \mathbf{x}^*) &= \mathbf{V}^T(\mathbf{x}^{[t]} - \mathbf{x}^*) - \alpha\mathbf{V}^T(2\mathbf{A}\mathbf{x}^{[t]} - \mathbf{b}) \\ \mathbf{w}^{[t+1]} &= \mathbf{w}^{[t]} - \alpha\mathbf{V}^T(2\mathbf{A}(\mathbf{x}^{[t]} - \mathbf{x}^*) + \underbrace{2\mathbf{A}\mathbf{x}^* - \mathbf{b}}_{=0}) \\ &= \mathbf{w}^{[t]} - 2\alpha\mathbf{\Lambda}\mathbf{V}^T(\mathbf{x}^{[t]} - \mathbf{x}^*) \\ &= \mathbf{w}^{[t]} - 2\alpha\mathbf{\Lambda}\mathbf{w}^{[t]}\end{aligned}$$



GD STEPS IN EIGENSPACE



Therefore, for each coordinate i :

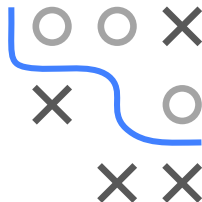
$$\begin{aligned}w_i^{[t+1]} &= w_i^{[t]} - 2\alpha\lambda_i w_i^{[t]} \\&= (1 - 2\alpha\lambda_i)w_i^{[t]} \\&= (1 - 2\alpha\lambda_i)^{t+1}w_i^{[0]}\end{aligned}$$

GD ERROR IN ORIGINAL SPACE

- Move back to the original space:

$$\mathbf{x}^{[t]} - \mathbf{x}^* = \mathbf{V}\mathbf{w}^{[t]} = \sum_{i=1}^d (1 - 2\alpha\lambda_i)^t w_i^{[0]} \mathbf{v}_i$$

- This gives the difference in $\mathbf{x}$ -space in closed form
- The difference vector is written in the eigenbasis, initial error components $w_i^{[0]}$ decay with rate $1 - 2\alpha\lambda_i$
- The smaller $|1 - 2\alpha\lambda_i|$ is, the faster component i decays; values close to 1 mean almost no progress per step
- Eigenvectors with larger eigenvalues typically converge fastest, especially early on



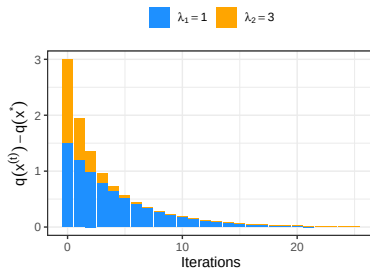
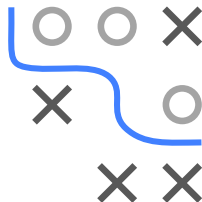
GD REGRET

- We can also analyze the regret (in objective value space):

$$q(\mathbf{x}^{[t]}) - q(\mathbf{x}^*) = \frac{1}{2} \sum_{i=1}^d (1 - 2\alpha\lambda_i)^{2t} \lambda_i (w_i^{[0]})^2$$

(straightforward to derive, formulas only become slightly lengthy)

- Rates become more similar once the slow modes (eigenvector components with small λ_i) dominate



OPTIMAL STEP SIZE

- Convergence requires $|1 - 2\alpha\lambda_i| < 1$ for all i
- Therefore $0 < \alpha\lambda_i < 1$
- The slowest component governs the rate:

$$\max_{i=1,\dots,n} |1 - 2\alpha\lambda_i| = \max(|1 - 2\alpha\lambda_1|, |1 - 2\alpha\lambda_n|)$$

- The optimum occurs at $\alpha = \frac{1}{\lambda_1 + \lambda_n}$
- The optimal rate equals $\frac{\lambda_n/\lambda_1 - 1}{\lambda_n/\lambda_1 + 1}$
- Convergence is driven by the condition number $\kappa = \frac{\lambda_n}{\lambda_1}$
- $\kappa = 1$ is ideal and yields convergence in one step

